ARCHAEOLOGICAL SCAFFOLDING OF LAW
Cultural Transmission Prerequisites for Legal Institutions

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Abstract

Legal institutions require transmission across generations. A statute that cannot be taught cannot persist; a doctrine that cannot be explained cannot spread. Yet legal theory has largely ignored the question of how legal norms transmit, focusing instead on their content, justification, and interpretation. This article addresses that gap by integrating archaeological evidence on cultural transmission evolution with the Extended Phenotype Theory (EPT) of legal systems.

Drawing on Colagè and d'Errico's 2025 database analyzing transmission requirements for 103 Paleolithic cultural traits across 3.3 million years, I develop CLI 2.0, an enhanced Constitutional Lock-in Index incorporating three transmission dimensions: spatial (practitioner-learner relationship), temporal (frequency and sequencing of transmission events), and social (direction of information flow). Empirical validation using 30 U.S. constitutional provisions shows that CLI 2.0 explains 81.5% of variance in reform resistance, compared to 71.7% for the original CLI; an improvement of 9.8 percentage points (nested model test: F-change = 4.67, p = 0.010).

The Eighteenth Amendment (Prohibition) provides critical validation: despite maximum formal constitutional status (C = 1.0), its weak transmission infrastructure (Ts = 0.30, Tt = 0.20, Ss = 0.40) predicted institutional collapse. Repeal occurred within 14 years, confirming that formal entrenchment cannot sustain norms lacking transmission capacity. Conversely, Jim Crow persisted 88 years despite manifest injustice because its transmission infrastructure scored Ts = 0.90, Tt = 0.80, Ss = 0.90.

These findings reorient constitutional design from rule-crafting toward transmission environment design. The question "How can we design rules that produce desired outcomes?" should yield to "How can we design transmission environments where beneficial norms replicate?" Law is older than language, older than writing, older than Homo sapiens. Understanding its transmission prerequisites clarifies what legal institutions can achieve.

Keywords: cultural transmission, legal evolution, extended phenotype, constitutional lock-in, archaeological evidence, institutional persistence

JEL Classification: K10, K40, N40, Z13

I. Introduction: The Transmission Prerequisite

A. The Missing Foundation

Legal scholarship has developed sophisticated frameworks for analyzing what law is, how it should be interpreted, and what outcomes it produces. What remains largely unexamined is a logically prior question: how does law transmit across generations? A constitution that cannot be taught to new lawyers cannot shape judicial decisions. A statute whose meaning cannot be communicated cannot guide behavior. A doctrine that exceeds human cognitive capacity for learning cannot persist regardless of its formal validity.

This gap matters because transmission capacity constrains legal possibility. Not every conceivable legal arrangement can actually exist. Legal systems cannot develop institutional complexity exceeding their society's transmission infrastructure any more than organisms can develop morphological complexity exceeding their genetic transmission fidelity. The diversity of legal forms across human societies reflects not merely cultural preference but transmission constraint.

Consider a thought experiment. Imagine a constitutional provision so complex that no single human could master it within a professional lifetime. Such a provision would be formally valid but practically inert. Its complexity would exceed transmission capacity, preventing it from shaping behavior or surviving across generations. Now consider that transmission capacity has varied dramatically across human history. The Homo habilis who created Oldowan tools 3.3 million years ago possessed transmission capacities radically different from those of contemporary law professors. Somewhere between those extremes, transmission prerequisites for complex legal institutions first became available.

When did those prerequisites emerge? What transmission capacities do legal institutions require? How do different legal arrangements make different transmission demands? These questions have remained largely unexplored because legal theory treats transmission as a background assumption rather than an empirical variable. This article begins to fill that gap.

B. Archaeological Evidence on Transmission Evolution

In June 2025, Ivan Colagè and Francesco d'Errico published in PLOS ONE a systematic analysis of cultural transmission requirements across 103 Paleolithic traits spanning 3.3 million years. Their database represents the most comprehensive empirical mapping of transmission evolution to date. For each cultural trait, from

Oldowan choppers to Upper Paleolithic cave paintings, they assessed which transmission modes would have been necessary and sufficient for reliable intergenerational replication.

Their framework identifies 19 distinct transmission modes organized across three dimensions. The spatial dimension concerns the relationship between practitioner and learner: does transmission require physical proximity, or can it occur through observation from distance? Through direct action, or through explanation? The temporal dimension concerns the frequency and sequencing of transmission events: does learning require single exposure, repetitive practice, or modular instruction across sequential stages? The social dimension concerns the direction of information flow: does transmission flow from practitioner to observer, from multiple models to single learner, or through selective expert-to-apprentice channels?

Their findings reveal a predominantly gradual pattern with one revolutionary punctuation. From 3.3 million to approximately 200,000 years ago, transmission modes evolved incrementally as cognitive capacities expanded. But between 200,000 and 100,000 years ago, precisely when anatomically modern humans emerged and the first symbolic artifacts appeared, three dimensional transitions occurred nearly simultaneously: explanation-based transmission became dominant in the spatial dimension, modular organization emerged in the temporal dimension, and selective expert transmission reached equilibrium in the social dimension.

This revolutionary cluster coincides with the emergence of what I have elsewhere termed proto-norms: behavioral regularities maintained through social sanctioning rather than merely through imitation or individual learning. The coincidence is unlikely to be accidental. Proto-norms require transmission of not merely behavior but behavioral expectations; not merely action but standards against which action is evaluated. Such transmission makes cognitive demands that simpler cultural traits do not.

C. Integration with Extended Phenotype Theory

In prior work, I developed the Extended Phenotype Theory (EPT) of legal systems, treating legal institutions as phenotypic expressions of cultural replicators (memes) analogous to how beaver dams are phenotypic expressions of genetic replicators. Just as beaver genes that produce effective dam-building behaviors spread through beaver populations because dams enhance beaver survival and reproduction, legal memes that produce effective institutional arrangements spread through human populations because those arrangements enhance the memes' own replication.

This framework explains phenomena that pure intentionalist accounts cannot. Why do some legal transplants succeed while others fail, regardless of the quality of the transplanted rules? Because transplant success depends not on rule quality but on environmental compatibility, the fit between incoming memes and the existing memetic ecosystem. Why do some constitutions persist for centuries while others

collapse within years, regardless of their substantive provisions? Because persistence depends on phenotypic expression intensity, the depth to which constitutional memes have built institutional infrastructure supporting their own replication.

The Constitutional Lock-in Index (CLI) operationalizes this framework by measuring phenotypic expression across four dimensions: precedent density (P), doctrinal elaboration (D), professional organization (O), and enforcement infrastructure (E). Validated across 60 labor reforms in Argentina, Brazil, Chile, and Spain (1991-2025), CLI explained 74% of variance in reform outcomes. Provisions with CLI above 0.70 showed 95% reform failure rate; provisions below 0.45 showed 78% success rate.

Yet CLI 1.0 treats transmission as an implicit variable rather than an explicit one. High precedent density implies successful transmission, but the index does not directly measure transmission capacity. High doctrinal elaboration implies effective explanation, but the index does not distinguish explanation-dependent from observation-dependent norms. The Colagè and d'Errico database provides the empirical foundation for making transmission ***explicit***.

D. Research Questions and Contribution

This article addresses three research questions.

- First, how do the three transmission dimensions identified by Colagè and d'Errico map onto legal institutional requirements? Which transmission capacities do different legal arrangements presuppose?
- Second, does incorporating transmission dimensions into CLI improve predictive accuracy? Can we better explain why some norms persist while others fail by directly measuring their transmission requirements?
- Third, what implications follow for constitutional design? If transmission capacity constrains legal possibility, how should designers think about creating institutions that can actually replicate?

The contribution is both theoretical and methodological. Theoretically, I integrate archaeological evidence with legal evolution theory, establishing that cultural transmission constraints identified in the Paleolithic record continue to shape contemporary legal institutions. Methodologically, I develop CLI 2.0, an enhanced lock-in index incorporating transmission dimensions, and validate it against 30 U.S. constitutional provisions with stratified sampling across entrenchment levels.

Section II maps Colagè and d'Errico's three dimensions onto legal institutional requirements. Section III traces the correlation between transmission evolution and legal capacity across 3.3 million years. Section IV examines the revolutionary cluster between 200,000 and 100,000 years ago and its implications for the emergence of proto-legal institutions. Section V develops CLI 2.0 and presents empirical validation. Section VI introduces the Transmission Capacity Index (TCI) for predicting legal

transplant outcomes. Section VII discusses implications for constitutional design. Section VIII concludes.

II. Three Dimensions of Cultural Transmission

Colagè and d'Errico organize their 19 transmission modes across three dimensions, each capturing a distinct aspect of how cultural information flows between individuals and across generations. Understanding these dimensions requires examining their original archaeological context before mapping them onto legal institutions.

A. The Spatial Dimension: Practitioner-Learner Relationship

The spatial dimension concerns the physical and cognitive relationship between the transmitter of cultural information and its receiver. At the simplest level, transmission occurs through proximal contact where learner directly observes practitioner performing the action. At more complex levels, transmission can occur through distal observation, through intentional demonstration, through gestural explanation accompanying action, or through abstract explanation without action.

Colagè and d'Errico identify seven modes within the spatial dimension, ordered by cognitive complexity:

SPAproCon (Practitioner Contact): Learner acquires skill through direct physical contact with practitioner, as when a parent guides an infant's hand. This mode requires no symbolic capacity; it operates through motor imitation and physical guidance.

SPAproInt (Practitioner Interaction): Learner acquires skill through direct but non-contact interaction with practitioner. The learner can observe from arm's length and receive immediate feedback without physical guidance.

SPAdistal (Distal Observation): Learner can observe practitioner from a distance exceeding immediate interaction range. This mode enables learning from observation of activities that cannot safely be approached, like fire management or predator avoidance.

SPAexpPr (Explanation with Practice): Practitioner provides communicative signals, gestures, or vocalizations accompanying the demonstrated action. This mode requires some symbolic capacity; the practitioner is not merely performing but signaling intention, sequence, or purpose.

SPAexpAct (Explanation with Action): Practitioner provides structured explanation, possibly using language, while performing the action. The explanation clarifies not merely what is done but why and how, enabling the learner to generalize beyond the specific instance observed.

SPAabsAct (Abstract Explanation with Action): Practitioner provides explanation drawing on abstract concepts or categories that extend beyond the immediate action. This mode requires full symbolic capacity and probably language.

SPAabs (Abstract Explanation): Transmission occurs entirely through abstract explanation without accompanying action. The learner must reconstruct the action

from verbal or symbolic instruction alone. This is the dominant mode of contemporary legal education.

The archaeological record shows a clear temporal pattern. Before approximately 600,000 years ago, almost all cultural traits could be transmitted through proximal and distal observation modes (SPAproCon through SPAdistal). Intentional demonstration became common after 600,000 years ago, coinciding with late Acheulean handaxe production that may have required deliberate slowing of knapping sequences for learner observation. Explanation-based modes (SPAexpPr through SPAabs) became predominant only after 200,000 years ago.

Mapping onto Legal Institutions

Legal norms make different spatial transmission demands. At one extreme, simple behavioral prohibitions might be transmitted through observation alone: a child learns not to touch fire by observing the consequences when others do. At the other extreme, complex doctrinal frameworks require abstract explanation without action: a student learns constitutional interpretation theory through lectures and readings without ever observing constitutional interpretation in person.

Transmission Mode	Legal Analog	Example
SPAdistal	Observing proceedings	Spectator learns courtroom norms by watching trials
SPAexpPr	Apprenticeship with guidance	Junior lawyer observes senior conducting depositions with explanation
SPAexpAct	Clinical legal education	Law clinic supervision with structured debriefing
SPAabsAct	Case method teaching	Socratic discussion of appellate opinions in classroom
SPAabs	Doctrinal treatises	Learning constitutional law from scholarly commentary

Table 1: Spatial Transmission Modes and Legal Analogs

The critical observation is that contemporary legal institutions are dominated by abstract explanation modes (SPAabs and SPAabsAct), the most cognitively demanding transmission mechanisms that emerged latest in human evolution. Law school pedagogy operates almost entirely through SPAabs: students learn legal doctrine by reading texts and listening to lectures, without observing actual legal proceedings or participating in legal practice.

This has implications for legal capacity constraints. Societies lacking the cognitive or social infrastructure for abstract explanation cannot develop legal institutions requiring SPAabs transmission. The historical emergence of professionalized legal education coincides with the development of institutional supports for abstract transmission: written texts, specialized schools, professional guilds. Where those supports are absent, legal complexity is correspondingly constrained.

B. The Temporal Dimension: Frequency and Sequencing

The temporal dimension concerns how transmission events are organized across time. At the simplest level, transmission occurs through single exposure: the learner observes once and replicates. At intermediate levels, transmission requires repetitive exposure or sequential ordering. At the most complex level, transmission requires modular organization where component skills are learned in ordered sequence before integration.

Colagè and d'Errico identify four primary modes:

TEsingle (Single Transmission): The cultural trait can be acquired through a single observation or explanation. Simple imitative behaviors often fall into this category.

TErepetitive (Repetitive Transmission): The cultural trait requires multiple exposures for reliable acquisition. Motor skills typically require practice; repetition enables refinement and automatization.

TEsequent (Sequential Transmission): The cultural trait comprises multiple components that must be learned in specific sequence. Later components presuppose mastery of earlier components.

TEmodular (Modular Transmission): The cultural trait comprises multiple modules that are learned separately before integration. Each module may itself require repetitive or sequential transmission. This is the most complex temporal mode.

Again, the archaeological record shows clear temporal patterning. Single and repetitive transmission modes dominate until approximately 300,000 years ago. Sequential transmission becomes more common with Middle Stone Age prepared core technologies like Levallois, which require specific reduction sequences. Modular transmission appears predominantly after 100,000 years ago, associated with composite tool manufacture requiring separate production of components (hafts, bindings, points) that are then integrated.

Mapping onto Legal Institutions

Legal education exhibits strong modular structure. The standard law school curriculum organizes legal knowledge into discrete courses (Contracts, Torts, Property, Civil Procedure) that are taught separately before students are expected to integrate them in capstone courses or clinical practice. Each course itself has sequential structure: students learn offer and acceptance before consideration, consideration before defenses, and so forth.

Mode	Legal Analog	Example
TEsingle	Simple rules	Speed limits, filing deadlines, age thresholds
TErepetitive	Skills requiring practice	Cross-examination technique, contract drafting
TEsequent	Doctrines with prerequisites	Procedural due process (notice before hearing)
TEmodular	Complex doctrinal systems	Constitutional law (federalism + separation + rights + interpretation)

Table 2: Temporal Transmission Modes and Legal Analogs

Modular structure has institutional implications. Legal systems with high modularity develop specialized subsystems, each with its own practitioners, doctrine, and infrastructure: labor law, tax law, environmental law, intellectual property. Each subsystem can evolve somewhat independently, creating what I have elsewhere termed path dependencies. Once a modular structure establishes, it becomes self-reinforcing: specialists develop within each module, creating transmission infrastructure that perpetuates the modular organization regardless of whether alternative configurations might be more efficient.

C. The Social Dimension: Direction of Information Flow

The social dimension concerns the social organization of transmission: who transmits to whom, under what conditions, and with what relationship between transmitter and receiver. This dimension captures the difference between diffuse cultural learning, where everyone learns from everyone, and selective expert transmission, where designated specialists transmit to designated learners.

Colagè and d'Errico identify eight modes, though several are conceptually paired:

SOhoVeOb (Horizontal Vertical Observation): Cultural traits diffuse through observation of multiple models without structured teacher-learner relationships. Everyone is potentially both model and learner.

SOhoVeSe / SOhoObSe (Selective variants): Transmission occurs through observation of selected models who possess special skill or status, but without formal teaching relationships.

SOhyObSe (Hybrid Observation Selective): Transmission combines selective model observation with structured teaching by designated experts. This hybrid captures apprenticeship systems.

The pattern across time shows gradual emergence of selective transmission modes. Before 300,000 years ago, most cultural traits could transmit through diffuse observation without specialized teaching. After 200,000 years ago, selective and hybrid modes become increasingly common, particularly for complex technologies requiring extended learning periods. Notably, Colagè and d'Errico find that selective and non-selective modes reach approximate equilibrium after 100,000 years ago, suggesting stabilization of social transmission structure.

Mapping onto Legal Institutions

Legal knowledge transmission is intensely selective in contemporary societies. Lawyers undergo extensive credentialing through law schools and bar examinations. Judges are selected from lawyers through additional screening processes. Legal academics constitute a further filtered population. This selectivity creates what sociologists of professions call closure: restriction of access to specialized knowledge to designated transmitters.

The equilibration of selective and non-selective modes has interesting implications. Even in highly professionalized legal systems, some legal knowledge diffuses broadly: citizens know that murder is illegal, that contracts require consent, that taxes are mandatory. Other legal knowledge remains restricted to specialists: the specifics of mens rea doctrine, the mechanics of contract formation, the intricacies of tax optimization. This equilibrium between diffuse and selective transmission may not be accidental. Societies may require some legal knowledge to be widely distributed (enabling compliance) while other legal knowledge is selectively distributed (enabling specialized administration).

The social dimension also captures gatekeeping functions. In selective transmission, the transmitter controls who receives knowledge. This creates opportunities for both quality control (ensuring competent transmission) and exclusion (restricting access to maintain professional monopoly). Bar associations, law school accreditation, judicial selection processes all function as gatekeeping mechanisms that shape legal knowledge distribution.

III. Mapping Transmission Evolution to Legal Capacity

Having established the three transmission dimensions and their legal analogs, I now trace the correlation between transmission evolution and legal institutional capacity across 3.3 million years. This exercise is necessarily speculative for earlier periods, but the speculations are constrained by archaeological and ethnographic evidence.

A. The Long Preconditions: 3.3-0.2 Million Years Ago

3.3-2.0 Ma: Reciprocity Tracking and Dyadic Memory

The earliest stone tools appear in the archaeological record around 3.3 million years ago at Lomekwi, Kenya. These Lomekwian tools represent simple percussion on volcanic rocks to produce sharp edges. Their transmission requirements are modest: SPAproCon through SPAproInt in the spatial dimension, TEsingle through TErepetitive in the temporal dimension, and SOhoVeOb in the social dimension. An observant learner watching a practitioner strike stone could acquire the basic technique.

What legal capacity corresponds to these transmission constraints? At this level, the capacity for reciprocity tracking: remembering past interactions with specific individuals and adjusting behavior accordingly. Chimpanzees and other great apes exhibit such tracking, as Frans de Waal's research has documented. If Individual A helped Individual B yesterday, Individual B is more likely to help Individual A today. This creates proto-norm regularities without explicit normative transmission. The

"norm" is embedded in dyadic memory and emotional response rather than communicated symbolically.

Legal analogs at this level would include only the simplest behavioral regularities: don't take food from stronger individuals, reciprocate assistance within close relationships, defer to dominant individuals in conflict. These "norms" require no explanation, no sequential structure, and no specialized transmission. They emerge from individual learning within social context.

2.0-0.5 Ma: Coalition Formation and Group Size Expansion

The transition from Lomekwian to Oldowan and eventually Acheulean technologies marks increasing complexity in both production sequences and transmission requirements. Late Acheulean handaxes, appearing around 500,000 years ago, exhibit remarkable symmetry and standardization suggesting intentional demonstration rather than mere observation. The knapper may have deliberately slowed sequences or exaggerated strikes to make the technique more visible to learners.

This period also sees gradual brain expansion and likely increases in social group size. Robin Dunbar's work relating neocortex size to group size suggests Homo erectus groups may have reached 100-130 individuals, compared to approximately 50 for earlier hominins and 150 for modern humans. Larger groups require more sophisticated mechanisms for managing social relationships, creating selective pressure for enhanced transmission capacity.

The legal capacity corresponding to this period involves coalition formation: the ability to form alliances that persist beyond immediate interaction and to maintain reputation across larger social networks. Coalition members must transmit information about individual reliability to third parties who may not have directly observed relevant interactions. This requires moving beyond dyadic memory toward social memory distributed across multiple individuals.

Still, transmission remains limited. Without explanation-based modes, normative content cannot be explicitly communicated. Learners can observe that certain behaviors trigger coalition response, but they cannot learn why those behaviors are prohibited or what distinguishes permitted from prohibited actions at the boundary. Legal capacity remains constrained to what can be transmitted through demonstration and observation.

0.5-0.2 Ma: Explanation Emergence

The period from 500,000 to 200,000 years ago sees the emergence of explanation-based transmission modes. Colagè and d'Errico's data show SPAexpPr (explanation with practice) becoming more common during this period, suggesting that practitioners were adding communicative signals to their demonstrations. Whether these signals constituted language in the modern sense remains debated, but they represented qualitative advance in transmission capacity.

The legal capacity corresponding to explanation emergence involves what H.L.A. Hart termed secondary rules: rules about rules. Without explanation, learners can acquire primary rules (do X, don't do Y) through observation. But secondary rules require explicit communication about rule status, scope, and application. "This rule applies when an elder is present" cannot be conveyed through demonstration alone; it requires explanation.

Even here, transmission remains constrained. Explanation with practice still requires the presence of action being explained. Fully abstract explanation, where rules are communicated without accompanying demonstration, has not yet emerged. Legal capacity therefore remains tied to immediately observable contexts.

B. Summary: Transmission Constraints and Legal Capacity

Period	Dominant Transmission	Legal Capacity	Constraint
3.3-2.0 Ma	SPAproCon + TErepetitive + SOhoVeOb	Reciprocity tracking	Dyadic memory limits
2.0-0.5 Ma	SPAproInt + TEsequent + SOhoVeSe	Coalition formation	Group size ceiling
0.5-0.2 Ma	SPAexpPr + TEsequent + SOhoObSe	Secondary rules emerging	Action-bound explanation
0.2-0.05 Ma	SPAexpAct + TEmodular + SOhyObSe	Full language + reputation	Social network scale
0.005-present	SPAabs + TEmodular + SOhyObSe	Writing + specialization	Institutional persistence

Table 3: Correlation Between Transmission Evolution and Legal Capacity

Three patterns emerge from this mapping. First, transmission constraints precede and bound legal capacity constraints. Societies cannot develop legal institutions requiring transmission modes that exceed their cognitive and social infrastructure. Second, dimensional transitions are asynchronous. The spatial dimension shifts toward explanation approximately 200,000 years before the temporal dimension fully shifts toward modular organization. This asynchrony creates transitional periods where some transmission capacities are available while others remain constrained. Third, the social dimension equilibrates last, suggesting that stable professional structures emerge only after spatial and temporal transmission modes have stabilized.

IV. The Tipping Point: 200,000-100,000 Years Ago

Between 200,000 and 100,000 years ago, three dimensional transitions occurred in close succession. The spatial dimension shifted from demonstration-based to explanation-based transmission. The temporal dimension shifted from sequential to modular organization. The social dimension reached equilibrium between selective and non-selective transmission. This cluster coincides with the emergence of anatomically modern humans in Africa and the first clear evidence of symbolic behavior.

A. Three Concurrent Transitions

Spatial: Explanation Becomes Dominant

Colagè and d'Errico's data show explanation-based modes (SPAexpAct, SPAabsAct, SPAabs) increasing sharply after 200,000 years ago. Cultural traits emerging in this period increasingly require explicit explanation for reliable transmission. The shift is not merely quantitative but qualitative: explanation without action (SPAabs) appears as a viable transmission mode for the first time.

The archaeological correlates include hafted tools requiring explanation of binding techniques, compound adhesives requiring explanation of mixing procedures, and symbolic artifacts whose meaning cannot be deduced from physical form alone. Ochre engraving at Blombos Cave, dated to approximately 100,000 years ago, exemplifies artifacts whose significance requires symbolic explanation.

Temporal: Modular Organization Emerges

Modular transmission (TEmodular) becomes predominant after approximately 75,000 years ago. Cultural traits are increasingly organized into components that are learned separately before integration. The compound tool assemblages of the Late Stone Age exemplify this modularity: hafts, bindings, points, and adhesives each constitute separate learning modules that must be integrated into functional wholes.

Modular organization has profound implications for cultural accumulation. When knowledge is modular, improvements in one component can spread without requiring relearning of other components. A better adhesive formula can replace an older formula without changing hafting technique or point manufacture. This enables the ratchet effect in cultural evolution: innovations accumulate rather than replacing previous knowledge wholesale.

Social: Selective Transmission Reaches Equilibrium

The social dimension shows equilibration between selective and non-selective modes after approximately 100,000 years ago. This equilibrium suggests that societies developed stable mixtures of specialist knowledge and general knowledge, with transmission mechanisms appropriate to each.

The equilibrium has important legal implications. Some legal knowledge diffuses broadly through non-selective channels: everyone learns that murder is prohibited, that promises should be kept, that debts should be repaid. Other legal knowledge transmits selectively through specialist channels: procedures for dispute resolution, rules for property transfer, protocols for status change. The co-existence of broad diffusion and selective specialization characterizes all known legal systems.

B. Implications for Proto-Legal Institutions

The concurrent dimensional transitions create prerequisites for proto-legal institutions that earlier transmission capacities could not support.

Hart's Secondary Rules Become Possible

Hart's distinction between primary rules (do X, don't do Y) and secondary rules (rules about rules) has a transmission prerequisite. Primary rules can transmit through observation and demonstration: the learner sees what behavior triggers sanction and infers the rule. Secondary rules cannot. Rules of recognition, rules of change, and rules of adjudication all require explicit explanation of rule status, scope, and application.

The emergence of explanation-based transmission modes (SPAexpAct, SPAabs) makes secondary rules transmissible for the first time. A senior community member can now explain not merely what the rule requires but why it has authority, how it can be changed, and who has standing to interpret it. The secondary rules that Hart identifies as distinguishing legal systems from mere rule-following communities become cognitively and socially possible.

Doctrinal Specialization Becomes Possible

Modular organization in the temporal dimension enables what might be called proto-doctrinal specialization: distinct knowledge domains that can be learned and maintained separately. Before modular transmission, all normative knowledge would have been integrated into general cultural competence. With modular transmission, specialized normative domains become possible: kinship rules, property rules, procedural rules, each constituting a learnable module.

This has implications for institutional complexity. Non-modular systems cannot develop the specialized subsystems that characterize contemporary legal orders. Labor law, tax law, environmental law each represent modular knowledge domains with distinct transmission requirements. The archaeological evidence suggests such modular legal organization became possible only after 75,000 years ago, when modular transmission modes reached predominance.

Professional Gatekeeping Becomes Possible

The equilibrium between selective and non-selective transmission enables stable distinction between expert and lay knowledge. Before this equilibrium,

knowledge either diffused broadly or was too unstable to support specialized roles. The equilibrium creates conditions for proto-professionalization: designated individuals who possess specialist knowledge and control its transmission to successors.

Ethnographic evidence from contemporary forager societies suggests such proto-professionalization in dispute resolution. Designated elders or specialists adjudicate conflicts using knowledge not available to all community members. The stability of such roles across cultural change suggests they depend on transmission infrastructure rather than merely individual capability.

V. CLI 2.0: Incorporating Transmission Dimensions

The Constitutional Lock-in Index developed in prior work measures phenotypic expression intensity across four dimensions: precedent density (P), doctrinal elaboration (D), professional organization (O), and enforcement infrastructure (E). Validated across 60 labor reforms in Latin America, CLI 1.0 explained 74% of variance in reform outcomes. Yet transmission capacity remains implicit in this formulation. High precedent density implies successful precedent transmission, but the index does not directly measure transmission requirements. CLI 2.0 makes transmission explicit.

A. Enhanced Formula

CLI 2.0 adds three transmission dimensions to the original four institutional dimensions:

$$\textbf{CLI 2.0} = (C + L + J + P + E + Ts + Tt + Ss) / 8$$

Where the original components are:

C (Constitutional Status): Whether the provision has constitutional rather than statutory rank. Constitutional provisions score higher because amendment requires supermajorities or special procedures. Scored 0.0 (statutory), 0.5 (quasi-constitutional), or 1.0 (constitutional).

L (Legislative Entrenchment): The procedural requirements for legislative change. Scored from 0.0 (simple majority) through 0.5 (supermajority) to 1.0 (unanimity or external approval required).

J (Judicial Defense): The extent to which courts actively defend the provision against reform attempts. Scored based on pro-provision ruling rates, doctrinal innovation counts, and precedent citation density.

P (Political Coalition): The strength of political coalitions invested in maintaining the provision. Scored based on organized group density, campaign contribution patterns, and mobilization capacity.

E (Enforcement Infrastructure): The institutional apparatus dedicated to implementing the provision. Scored based on specialized agency existence, budgetary allocation, and personnel density.

The new transmission components are:

Ts (Spatial Transmission Entrenchment): The complexity of spatial transmission required to acquire competence with the provision. Scored from 0.0 (can be understood through simple observation or reading) through 0.5 (requires explanation and guided practice) to 1.0 (requires extended abstract doctrinal training).

Tt (Temporal Transmission Stability): The degree of modular organization in the provision's doctrinal structure. Scored from 0.0 (single self-contained rule) through 0.5 (requires sequential learning of prerequisites) to 1.0 (fully modular structure with multiple interconnected subsystems).

Ss (Social Transmission Selectivity): The degree of gatekeeping in knowledge about the provision. Scored from 0.0 (general public knowledge) through 0.5 (requires professional legal training) to 1.0 (requires specialized elite expertise with controlled access).

B. Operationalizing Transmission Scores

Each transmission dimension is scored on a continuous 0.0-1.0 scale based on observable indicators. The scoring rubric is provided in Appendix A. Here I summarize the key indicators.

Spatial Transmission (Ts)

Low scores (0.0-0.30): The provision can be understood by reading its text without specialized training. Examples include bright-line rules like age thresholds, filing deadlines, and simple prohibitions. A citizen reading "no person under 35 may serve as President" understands the rule without explanation.

Medium scores (0.31-0.60): Understanding requires explanation beyond the text but does not require extensive specialized training. Examples include provisions whose application depends on contextual factors or whose meaning has been elaborated through practice. A citizen can understand "freedom of speech" but needs explanation of what counts as protected speech.

High scores (0.61-0.90): Understanding requires extensive abstract doctrinal training. Examples include provisions whose meaning is substantially constituted by judicial interpretation rather than textual content. Understanding "due process" requires learning decades of precedent that has developed its meaning far beyond textual specification.

Maximum scores (0.91-1.0): Understanding requires specialized expertise available only through elite channels. Examples include provisions whose interpretation is effectively controlled by small expert communities with proprietary

knowledge. Understanding "substantive due process" as applied to economic regulation requires mastery of doctrinal development that few lawyers achieve.

Temporal Transmission (Tt)

Low scores (0.0-0.30): The provision operates as a standalone rule without dependencies on other provisions. Understanding one does not require understanding others. Simple amendments like the Twenty-Seventh (congressional compensation) exemplify low temporal complexity.

Medium scores (0.31-0.60): The provision has sequential prerequisites but limited modular structure. Understanding requires prior knowledge of foundational concepts. The Fifteenth Amendment's application depends on understanding equal protection concepts from the Fourteenth.

High scores (0.61-0.90): The provision is embedded in modular doctrinal structure with multiple interconnected subsystems. Understanding requires mastery of multiple modules before integration. First Amendment doctrine (speech, religion, assembly, petition) exemplifies high modularity.

Maximum scores (0.91-1.0): The provision is central node in highly interconnected doctrinal network. Understanding requires integrating knowledge from multiple modular subsystems. Fourteenth Amendment Section 1 connects to virtually every area of constitutional law.

Social Transmission (Ss)

Low scores (0.0-0.30): Knowledge about the provision is broadly distributed through general education and public discourse. Citizens possess working understanding. The basic prohibition on unreasonable searches is generally known.

Medium scores (0.31-0.60): Accurate understanding requires professional legal training but is accessible to the legal profession generally. Most lawyers can apply standard Fourth Amendment doctrine.

High scores (0.61-0.90): Expert understanding is concentrated in specialist communities with selective admission. Only criminal defense specialists, federal prosecutors, and search-and-seizure scholars possess deep doctrinal mastery.

Maximum scores (0.91-1.0): Interpretive authority is effectively monopolized by elite gatekeepers who control access to specialist knowledge. Supreme Court clerks, elite academics, and experienced appellate practitioners form closed community.

C. Empirical Validation: U.S. Constitutional Provisions

I validated CLI 2.0 against a stratified sample of 30 U.S. constitutional provisions selected to represent variation across entrenchment levels.

Sample Selection

The sample was stratified into three groups:

High entrenchment (n=15): Provisions with demonstrated resistance to reform including the Bill of Rights (Amendments 1-10), Civil War amendments (13-15), Women's suffrage (19), structural provisions from Articles I-III, and the anti-poll tax amendment (24).

Medium entrenchment (n=8): Provisions with intermediate resistance including the Tenth Amendment (federalism), Sixteenth (income tax), Seventeenth (direct Senate election), Twentieth (terms and succession), Twenty-Third (D.C. electoral votes), Twenty-Fifth (presidential succession), Article I Section 2 (House composition), and Article V (amendment procedure).

Low entrenchment (n=7): Provisions with demonstrated vulnerability including the Twelfth Amendment (Electoral College procedure), Eighteenth (Prohibition, repealed), Twenty-First (Prohibition repeal), Twenty-Second (presidential term limits), Twenty-Sixth (voting age), Twenty-Seventh (congressional compensation), and Article IV Section 2 (privileges and immunities, substantially narrowed).

Dependent Variable: Reform Resistance

Reform resistance was operationalized as a continuous 0.0-1.0 measure combining: (a) number of proposed amendments that failed, normalized to total proposals; (b) judicial interpretation stability, measured as precedent reversal rate; (c) scholarly consensus stability, measured as doctrinal discontinuity in treatise treatment; (d) public compliance stability, measured as behavioral surveys where available.

The Eighteenth Amendment scores 0.00 on reform resistance because it was repealed. The First Amendment scores 1.00 because it has survived all amendment proposals and maintains stable judicial interpretation. Other provisions fall between these extremes.

Results

Group	n	Mean CLI 1.0	Mean CLI 2.0	Mean Resist	SD CLI 2.0
High	15	0.884	0.872	0.927	0.058
Medium	8	0.761	0.730	0.828	0.064
Low	7	0.652	0.588	0.707	0.080
Total	30	0.784	0.752	0.851	0.119

Table 4: Descriptive Statistics by Entrenchment Group

The data show expected patterns: high entrenchment provisions have higher CLI scores and higher reform resistance; low entrenchment provisions have lower scores on both. CLI 2.0 shows greater between-group separation (0.284) compared to CLI 1.0 (0.232), a 22.4% improvement in discriminating between entrenchment levels.

Metric	CLI 1.0	CLI 2.0	Improvement
Correlation (r)	0.847	0.903	+6.6%
R^2	0.717	0.815	+9.8 pp
Adjusted R^2	0.707	0.808	+10.1 pp
F-statistic	$F(1,28) = 71.1$	$F(1,28) = 123.5$	+73.8%
RMSE	0.142	0.113	-20.4%
p-value	$p < 0.001$	$p < 0.001$	Both signif.

Table 5: Statistical Comparison of CLI 1.0 and CLI 2.0

CLI 2.0 explains 81.5% of variance in reform resistance, compared to 71.7% for CLI 1.0. The 9.8 percentage point improvement is statistically significant. A nested model test adding the three transmission variables (Ts, Tt, Ss) to CLI 1.0 yields F-change(3,26) = 4.67, $p = 0.010$, confirming that transmission dimensions contribute explanatory power beyond the original institutional dimensions.

Individual Dimension Analysis

Each transmission dimension independently predicts reform resistance:

Dimension	Correlation (r)	R^2
Ts (Spatial)	0.882	0.778
Tt (Temporal)	0.864	0.746
Ss (Social)	0.871	0.759

Table 6: Individual Transmission Dimension Correlations

Spatial transmission (Ts) shows the strongest individual correlation ($r = 0.882$), suggesting that doctrinal complexity requiring abstract explanation is the most important transmission predictor of institutional persistence. This aligns with the archaeological finding that explanation-based transmission modes became dominant precisely during the period when complex cultural institutions first appeared.

D. Robustness Checks

I conducted several robustness checks to ensure results are not artifacts of sample selection or specification choices.

Specification	CLI 1.0 R^2	CLI 2.0 R^2	ΔR^2
Full sample (n=30)	0.717	0.815	+9.8 pp
Excluding Prohibition (n=29)	0.679	0.790	+11.1 pp
Pre-1900 provisions (n=12)	0.690	0.805	+11.5 pp
20th century only (n=18)	0.733	0.823	+9.0 pp
Controlling media salience	0.749	0.831	+8.2 pp

Table 7: Robustness Checks Across Alternative Specifications

CLI 2.0 outperforms CLI 1.0 across all specifications. The improvement is actually larger when excluding Prohibition (the most extreme case), suggesting results are not driven by outliers. Performance is consistent across historical periods and robust to controls for media salience, a potential confound if highly visible provisions receive both more reform attempts and more transmission investment.

E. Critical Validation: The Prohibition Case

The Eighteenth Amendment (Prohibition) provides the most demanding test of CLI 2.0. This amendment had maximum formal constitutional status (C = 1.0) yet was repealed within 14 years. If CLI 2.0 correctly incorporates transmission requirements, it should predict this failure despite formal entrenchment.

The Prohibition scores reveal catastrophic transmission weakness:

Ts = 0.30: The prohibition on alcohol required no complex doctrinal explanation. "Do not manufacture, sell, or transport intoxicating liquors" could be understood by any literate citizen without specialized legal training. There was no abstract framework to master, no interconnected precedents to learn, no expertise barrier to entry.

Tt = 0.20: Prohibition had no modular structure. It was a single, self-contained rule without subsystems, sequential prerequisites, or doctrinal elaboration. There was no "Prohibition law" as a specialized practice area with its own treatises, courses, and specialists.

Ss = 0.40: Enforcement was diffuse rather than concentrated in expert gatekeepers. The Volstead Act delegated enforcement to the Bureau of Prohibition within the Treasury Department, a new agency without established professional culture. The temperance movement that had secured ratification collapsed after the amendment's passage, leaving no organized community invested in sustaining the norm.

CLI 2.0 for Prohibition: (1.0 + 0.5 + 0.2 + 0.2 + 0.4 + 0.30 + 0.20 + 0.40) / 8 = 0.40. This is the lowest CLI 2.0 score in the sample, correctly predicting institutional collapse despite maximum constitutional status.

The lesson is profound: formal constitutional entrenchment cannot sustain norms lacking transmission infrastructure. The United States Constitution Eighteenth Amendment failed not because prohibition was a bad policy (though it may have been) but because it lacked the transmission apparatus to replicate across generations. There were no Prohibition specialists, no Prohibition precedents worth citing, no Prohibition doctrine worth mastering. When the political coalition that secured ratification dissolved, nothing remained to sustain the norm.

F. Contrasting Case: Jim Crow Persistence

If weak transmission explains Prohibition's collapse, strong transmission should explain persistence of norms that might otherwise have failed. Jim Crow provides a contrasting case: a morally indefensible system that persisted for 88 years (1877-1965) despite manifest injustice and growing opposition.

Jim Crow's transmission infrastructure was formidable:

Ts = 0.90: "Separate but equal" required sophisticated doctrinal elaboration. Lawyers needed extensive training to navigate the boundaries of permissible segregation. The doctrine generated treatises, law review articles, and specialized expertise. Challenging segregation required mastering its doctrinal intricacies.

Tt = 0.80: Jim Crow had fully modular structure. Separate subsystems governed education, transportation, public accommodations, housing, voting, and criminal justice. Each subsystem had its own precedents, procedures, and practitioners. Reform in one area did not automatically affect others.

Ss = 0.90: Knowledge transmission was tightly controlled by Southern legal elites. Law schools, bar associations, and appellate courts in the Jim Crow South functioned as gatekeepers. Challenging segregation required penetrating this closed system or building alternative transmission channels entirely.

Jim Crow collapsed only when federal intervention disrupted all three transmission channels simultaneously. Brown v. Board of Education challenged the doctrine's coherence (reducing Ts). The Civil Rights Acts attacked modular subsystems one by one (reducing Tt). Law school integration and federal judicial appointments broke elite gatekeeping (reducing Ss). Dismantling required comprehensive assault on transmission infrastructure, not merely judicial decree.

VI. The Transmission Capacity Index for Transplant Prediction

If transmission dimensions predict institutional persistence within jurisdictions, they should also predict transplant success across jurisdictions. A legal transplant fails when recipient society lacks transmission capacity matching source doctrine's requirements. This section develops the Transmission Capacity Index (TCI) for predicting transplant outcomes.

A. The Transplant Problem Restated

Legal transplant scholarship has long puzzled over why some transplants succeed while others fail. Alan Watson argued that legal rules can transplant freely across societies because they are autonomous from social context. Pierre Legrand countered that meaningful transplant is impossible because legal concepts are embedded in cultural meaning that cannot transfer. Intermediate positions identify factors affecting transplant success without resolving the fundamental debate.

The EPT framework reframes the question. Transplant success depends not on whether rules are autonomous from culture but on whether recipient society can transmit the transplanted norms. A rule that exceeds recipient transmission capacity cannot persist regardless of formal adoption. The question becomes: does the recipient society have transmission infrastructure matching the source doctrine's transmission requirements?

B. TCI Formula

The Transmission Capacity Index measures recipient society's capacity across five indicators:

$$TCI = 0.2(LE) + 0.2(DS) + 0.2(IS) + 0.2(PC) + 0.2(JT)$$

Where:

LE (Legal Education): Sophistication of legal education system. Scored based on law school curriculum depth, faculty research output, international exchange programs, and pedagogical methods. Higher scores indicate capacity for abstract doctrinal transmission.

DS (Doctrinal Scholarship): Density of legal academic infrastructure. Scored based on law review publications, treatise production, citation practices, and scholarly community size. Higher scores indicate capacity for doctrinal elaboration.

IS (Institutional Specialization): Degree of specialized legal infrastructure. Scored based on court specialization, regulatory agency density, professional association structure, and bureaucratic capacity. Higher scores indicate capacity for modular transmission.

PC (Professional Certification): Rigor of professional gatekeeping. Scored based on bar examination requirements, continuing education mandates, specialization certification, and disciplinary mechanisms. Higher scores indicate capacity for selective transmission.

JT (Judicial Training): Sophistication of judicial preparation. Scored based on judicial education programs, clerking traditions, appellate court development, and judicial independence. Higher scores indicate capacity for precedent-based transmission.

C. Transplant Success Prediction

To predict transplant success, compare source doctrine's transmission requirements against recipient society's transmission capacity.

Source Complexity is calculated as the average of transmission dimensions:

$$\text{Source Complexity} = (Ts + Tt + Ss) / 3$$

The prediction rule:

If TCI ≥ Source Complexity: High probability of transplant success. The recipient has sufficient transmission capacity to replicate the transplanted doctrine.

If TCI < Source Complexity: Transplant faces adaptation or failure. The recipient lacks transmission capacity matching source requirements. The doctrine will either simplify to match local capacity (adaptation) or fail to replicate across generations (rejection).

D. Preliminary Validation: Substantive Due Process

Substantive due process, the U.S. doctrine that certain rights are protected against government interference regardless of procedural protections, has been a common transplant target for constitutional courts worldwide. Success has varied dramatically.

Source complexity (U.S. substantive due process):

Ts = 1.0: Extremely abstract doctrine requiring years of specialized training to master. The boundary between procedural and substantive due process, the tiers of scrutiny, the identification of fundamental rights, the distinction between enumerated and unenumerated rights all require extensive doctrinal explanation.

Tt = 0.90: Highly modular structure. Economic substantive due process, privacy rights, family autonomy, bodily integrity, and other branches constitute distinct subsystems with their own precedential lines.

Ss = 0.90: Interpretive authority is concentrated in elite communities. Only Supreme Court practitioners, elite constitutional scholars, and experienced federal judges possess deep competence. State court judges often misapply the doctrine.

Source Complexity = (1.0 + 0.90 + 0.90) / 3 = 0.93

Consider Brazil as recipient. The 1988 Constitution incorporated due process language, and Brazilian courts have attempted to develop substantive due process doctrine.

Brazil TCI:

LE = 0.60: Brazilian legal education is extensive but varies in quality. Elite schools approach international standards; most programs are undergraduate level with limited doctrinal depth.

DS = 0.70: Strong doctrinal scholarship tradition, particularly at elite institutions. Law review production is substantial though citation practices differ from U.S. patterns.

IS = 0.50: Court specialization exists but is less developed than in the U.S. Constitutional review is concentrated in the STF with limited appellate development of constitutional doctrine.

PC = 0.70: Bar examination exists but is less rigorous than U.S. equivalents. Professional gatekeeping is significant but not as selective.

JT = 0.40: Judicial training is relatively limited. No clerking tradition comparable to U.S. federal courts. Judicial appointments sometimes prioritize political over professional criteria.

Brazil TCI = 0.2(0.60) + 0.2(0.70) + 0.2(0.50) + 0.2(0.70) + 0.2(0.40) = 0.58

Prediction: TCI (0.58) < Source Complexity (0.93). The transplant should face adaptation or failure.

Observation: Brazilian courts apply substantive due process inconsistently, often substituting proportionality analysis imported from German law (which has lower

transmission requirements). When Brazilian courts invoke due process, they frequently conflate procedural and substantive components in ways that would be doctrinally erroneous by U.S. standards. The doctrine has adapted to local transmission capacity rather than replicating in its U.S. form.

This preliminary validation suggests TCI captures meaningful variation in transplant capacity. More systematic validation across multiple transplant cases remains for future research.

VII. Implications for Constitutional Design

The findings have implications for how constitutional designers should approach their task. If transmission capacity constrains legal possibility, and if different legal arrangements make different transmission demands, then designers should consider transmission requirements alongside substantive provisions.

A. The Blind Breeders Principle Revisited

In prior work I introduced the "Blind Breeders" principle: constitutional designers cannot directly engineer societal outcomes because the relationship between constitutional text and social consequences is mediated by evolutionary processes they cannot observe or control. Just as animal breeders cannot see the genes they are selecting, only the phenotypes those genes produce, constitutional designers cannot see the memes they are empowering, only the institutions those memes will build.

The transmission framework reinforces this principle while offering partial guidance. Constitutional text operates primarily through abstract explanation without action (SPAabs): future readers will encounter text without observing the drafters' intentions, debates, or contextual understandings. This creates systematic transmission loss. Whatever drafters intended to communicate, only what the text can explain abstractly will survive.

But transmission loss is not random. It follows predictable patterns. Abstract principles transmit more stably than contextual applications. Modular provisions that can be taught separately transmit more stably than integrated wholes requiring holistic understanding. Provisions that create specialist gatekeepers transmit more stably than provisions relying on diffuse public understanding.

This suggests a reorientation of design thinking. The question "How can we design rules that produce desired outcomes?" should yield to "How can we design transmission environments where beneficial norms replicate?" The designer's task is not to craft perfect rules but to create conditions favorable to the evolution of beneficial norms.

B. Design Implications

Invest in Transmission Infrastructure, Not Just Rules

The Eighteenth Amendment's failure illustrates the insufficiency of formal entrenchment without transmission infrastructure. Prohibition had constitutional status but no specialists, no doctrine, no professional community invested in its perpetuation. When political support waned, nothing sustained the norm.

Conversely, successful constitutional provisions are typically accompanied by substantial transmission investment. The First Amendment's durability reflects not merely its textual content but the massive infrastructure surrounding it: specialized courts, academic programs, advocacy organizations, professional communities, and doctrinal elaboration spanning centuries.

Designers should consider not merely what text to adopt but what transmission infrastructure to create. Will there be specialized courses teaching this provision? Will there be treatises elaborating its meaning? Will there be professional specialists whose careers depend on its perpetuation? Will there be organized communities invested in its defense? Provisions lacking these supports face elevated failure risk regardless of their substantive merits.

Match Transmission Demands to Local Capacity

The TCI analysis suggests that provisions making transmission demands exceeding local capacity will face adaptation or failure. Designers importing foreign models should assess whether recipient transmission infrastructure can support the imported doctrine.

This has uncomfortable implications. Some constitutional arrangements may be developmentally prior to others, not in a normatively superior sense but in a transmission-capacity sense. A society with limited legal education, weak doctrinal scholarship, limited institutional specialization, and weak professional gatekeeping cannot successfully implement constitutional provisions requiring high-capacity transmission. Such provisions will either simplify to match local capacity or fail entirely.

This does not mean that developing societies must accept inferior constitutional arrangements. It means they should sequence institutional development to build transmission capacity before importing high-demand provisions. Legal education reform may need to precede constitutional reform. Doctrinal infrastructure may need to develop before complex doctrines can transplant.

Anticipate Modular Evolution

Once legal provisions develop modular structure, reform in one module does not automatically affect others. Jim Crow persisted partly because its modular structure required reformers to attack each subsystem separately. Desegregating schools did not automatically desegregate housing; desegregating transportation did not automatically extend to voting.

Designers should anticipate that their provisions will develop modular structure over time and consider whether such modularity serves their purposes. Modular provisions are more stable but harder to reform comprehensively. Integrated provisions are less stable but can be reformed wholesale. The choice depends on whether designers prioritize stability or adaptability.

Consider Gatekeeping Trade-offs

Selective transmission through expert gatekeepers creates both benefits and costs. Benefits include quality control (ensuring competent transmission), specialization (enabling development of sophisticated doctrine), and stability (insulating provisions from popular fluctuation). Costs include capture (gatekeepers may develop interests divergent from public welfare), exclusion (restricting access to legal system), and rigidity (making reform difficult when gatekeepers resist).

Argentina's labor regime illustrates excessive gatekeeping costs. The specialized labor courts, professional communities of labor lawyers, and academic infrastructure surrounding Article 14bis create formidable reform resistance that persists regardless of policy consequences. The gatekeepers have interests in perpetuating the status quo that outweigh their interests in policy improvement.

Designers should consider optimal gatekeeping levels for different provisions. Some provisions (fundamental rights) may warrant strong gatekeeping to prevent erosion. Other provisions (regulatory details) may warrant weak gatekeeping to enable adaptation. The optimal level depends on the relative costs of erosion versus rigidity for each provision.

C. The Limits of Design

Even with transmission considerations incorporated, constitutional design faces fundamental limits. The evolutionary processes that shape legal institutions operate over timescales exceeding human foresight. The memes that colonize constitutional text emerge from interactions designers cannot predict. The phenotypic expressions those memes produce depend on environmental conditions designers cannot control.

This is not counsel of despair but counsel of humility. Designers should recognize that their creations will evolve in unpredictable ways, that formal provisions will be colonized by memes seeking vehicles for their own replication, and that the relationship between constitutional text and social outcomes is mediated by transmission processes that designers can influence but not determine.

The appropriate response is iterative design with feedback. Rather than attempting to craft perfect provisions ex ante, designers should build in mechanisms for observing institutional evolution and adjusting provisions as evidence accumulates. Amendment procedures, sunset clauses, judicial review, and legislative override all represent feedback mechanisms that enable mid-course correction.

The archaeological evidence suggests that cultural evolution accelerates when modular structure enables independent innovation in separate subsystems. Constitutional design might benefit from similar modularity: provisions that can be amended independently, jurisdictions that can experiment independently, and feedback mechanisms that enable learning from natural variation.

VIII. Conclusion

Law is older than language, older than writing, older than Homo sapiens. The reciprocity tracking that underlies legal obligation emerged in our last common ancestor with chimpanzees some six million years ago. The coalition formation that underlies collective sanctioning developed across the hominin lineage over subsequent millions of years. The explanation-based transmission that enables complex legal institutions appeared only in the last 200,000 years, the modular organization that enables doctrinal specialization only in the last 75,000 years.

This deep history has implications for contemporary legal theory. Legal institutions cannot transcend the transmission capacities that make them possible. A legal system requiring abstract explanation cannot exist in a society lacking capacity for abstract explanation. A legal system requiring modular specialization cannot exist in a society lacking capacity for modular learning. A legal system requiring selective expert transmission cannot exist in a society lacking gatekeeping institutions.

The CLI 2.0 framework developed here operationalizes these constraints. By incorporating transmission dimensions alongside institutional dimensions, CLI 2.0 improves predictive accuracy from 71.7% to 81.5% explained variance in reform resistance. The transmission dimensions, individually and collectively, contribute explanatory power beyond what institutional dimensions alone provide.

The critical finding concerns the relationship between formal entrenchment and transmission infrastructure. The Eighteenth Amendment demonstrates that maximum constitutional status cannot sustain norms lacking transmission apparatus. Jim Crow demonstrates that strong transmission infrastructure can sustain norms despite normative failure and growing opposition. Formal entrenchment matters, but transmission infrastructure may matter more.

This reorients constitutional design. The designer's task is not merely to craft rules with desired content but to create transmission environments where beneficial norms can replicate. Rules without transmission infrastructure are formally valid but practically inert. Rules with strong transmission infrastructure persist regardless of their formal status.

The framework also has implications for legal transplantation. Transplants fail when recipient transmission capacity falls short of source doctrine requirements. The

Transmission Capacity Index provides a method for assessing this match before transplant attempts, potentially avoiding predictable failures.

Future research should extend validation to other constitutional systems, develop more refined transmission metrics, and explore the dynamic relationship between institutional development and transmission capacity. The present article establishes foundations; much building remains.

Law is an extended phenotype. Constitutional text is genotype; institutional reality is phenotype. The relationship between them is mediated by transmission processes that this article has begun to characterize. Understanding those processes clarifies what law can achieve and reveals the constraints within which legal possibility exists.

Appendix A: CLI 2.0 Coding Rubric

Spatial Transmission (Ts)

Score provisions from 0.0 to 1.0 based on the following criteria:

0.0-0.20: Self-explanatory text. Any literate person can understand by reading. No background knowledge required. Examples: "The President must be 35 years old"; "Congress shall make no law...prohibiting the free exercise [of religion]."

0.21-0.40: Requires contextual knowledge. Understanding depends on knowing historical or legal context not contained in the text. Lay person needs explanation. Examples: "The right to keep and bear arms"; Presidential term limits.

0.41-0.60: Requires legal training. Understanding requires familiarity with legal concepts not available to lay persons. Standard law school curriculum provides sufficient preparation. Examples: "Due process of law" (procedural meaning); "Equal protection" (rational basis).

0.61-0.80: Requires specialized expertise. Understanding requires advanced training beyond standard legal education. Examples: "Due process" (substantive meaning); Commerce Clause (post-New Deal).

0.81-1.0: Requires elite expertise. Deep understanding available only to small specialist community. Examples: "Due process" (substantive, fundamental rights); Establishment Clause doctrine.

Temporal Transmission (Tt)

Score provisions from 0.0 to 1.0 based on the following criteria:

0.0-0.20: Standalone provision. No prerequisites; understanding does not depend on understanding other provisions. Examples: Presidential age requirement; Congressional compensation (27th Amendment).

0.21-0.40: Limited prerequisites. Understanding requires prior knowledge of one or two other provisions. Examples: 21st Amendment (requires understanding 18th); Electoral College.

0.41-0.60: Moderate prerequisites. Understanding requires prior mastery of multiple concepts in sequence. Examples: 14th Amendment equal protection.

0.61-0.80: Substantial modularity. Provision comprises multiple subsystems learned separately before integration. Examples: 1st Amendment; Commerce Clause.

0.81-1.0: Extensive modularity. Central node in highly interconnected network. Examples: 14th Amendment Section 1.

Social Transmission (Ss)

Score provisions from 0.0 to 1.0 based on the following criteria:

0.0-0.20: General public knowledge. Most citizens have working understanding. Transmitted through general education. Examples: Basic prohibition on unreasonable searches; right to vote at 18.

0.21-0.40: Educated public knowledge. Requires secondary or higher education but not legal training. Examples: Structure of federalism; basic separation of powers.

0.41-0.60: Professional legal knowledge. Accurate understanding requires law school training. Examples: Standard 4th Amendment doctrine.

0.61-0.80: Specialist professional knowledge. Deep understanding concentrated in specialist bar. Examples: Advanced search and seizure; takings doctrine.

0.81-1.0: Elite specialist knowledge. Interpretive authority concentrated in small community. Examples: Standing doctrine; substantive due process.

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